

NITIN DUMKA

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PROFESSIONAL SUMMARY

AI/ML Engineer who improved retrieval accuracy by 30% and reduced hallucination by 25% through production-grade RAG systems. Experienced in building scalable LLM pipelines and optimizing semantic retrieval using LangChain, FAISS, and PyTorch. Strong in designing high-performance ETL workflows and deploying modular AI systems for real-world applications.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++

ML / AI: PyTorch, TensorFlow, scikit-learn, OpenCV, Deep Learning, NLP, XGBoost, Random Forest

LLM & Retrieval: LangChain, Ollama, Hugging Face Transformers, FAISS, ChromaDB, Semantic Search, Prompt Engineering

Data & Cloud: Pandas, NumPy, ETL Pipelines, AWS, MySQL, MongoDB

Tools: Docker, FastAPI, Streamlit, Git

PROFESSIONAL EXPERIENCE

Conversational Data Analysis Intern — VOIS, Vodafone Idea Foundation, Remote

Sep 2025 – Oct 2025

- Engineered production-grade RAG pipelines using LangChain, FAISS, and Ollama, improving answer relevance by 30% across large conversational datasets
- Designed modular retrieval architecture integrating embedding, indexing, and response synthesis for scalable deployment
- Optimized semantic chunking and embedding strategies, improving multi-turn retrieval precision and reducing hallucination in LLM responses

ESG Data Science Intern — Infosys Springboard, Remote

Aug 2025 – Oct 2025

- Developed ensemble ML models (Random Forest, XGBoost) on ESG datasets, improving stakeholder engagement by 18% using SHAP-based interpretability
- Automated end-to-end ETL pipelines in Python and SQL, reducing manual processing by 25% and enabling near real-time analytics

Remote Sensing ML Trainee — ISRO IIRS Dehradun

Sep 2025 – Oct 2025

- Implemented ML-based classification on multispectral satellite imagery, achieving 85% accuracy across 10,000+ sq km using high-dimensional geospatial data processing

AI/ML Engineer & Developer — Northrop I/O Technologies

Jul 2024 – Nov 2024

- Built a graph-based hospital recommendation system using Dijkstra's algorithm for optimized real-time routing decisions under constraints
- Developed AI-powered full-stack applications, improving system performance and reducing page load time by 25%

PROJECTS

Agentic Search & Query Decomposition (Ongoing)

- Designed plan-then-retrieve LLM architecture decomposing complex queries into 3-5 atomic sub-queries for independent retrieval and synthesis
- Reduced hallucination by ~25% on multi-hop reasoning benchmarks compared to single-turn dense_retrieval

LLM-Based Document QA System — RAG

- Built end-to-end RAG pipeline using LangChain, FAISS, and Ollama over multi-document corpora
- Optimized chunking strategies, embedding selection, and re-ranking to improve answer grounding and retrieval precision

AI Healthcare Recommendation System

- Developed interpretable ML system enabling explainable predictions for personalized healthcare recommendations

EDUCATION

B.Tech, Computer Science Engineering — Graphic Era Hill University

Expected May 2027

Coursework : AI, Machine Learning, Statistical Analysis, Linear Algebra, Probability & Statistics, DSA, DBMS

Certifications : Advances in Remote Sensing — ISRO (2025) | Generative AI & Prompt Engineering — Infosys (2025) | AWS Cloud Practitioner | Scientific Computing with Python — freeCodeCamp (2025)

ACHIEVEMENTS

- Ranked Top 5% in Excelerate Internship Program; Finalist at HackForge 2025 national-level hackathon
- Solved 350+ DSA problems across LeetCode and GeeksforGeeks